

STT 861: Theory of Probability & Statistics I

Lecture 13: Differentiating Under an Integral

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Oct 8, 2026 — Wells Hall C506

Reference: Casella & Berger Ch. 2.4

Why Do We Need to Differentiate Under an Integral?

The core problem

We repeatedly want to do this:

$$\frac{d}{d\theta} \int_{-\infty}^{\infty} f(x, \theta) dx \stackrel{?}{=} \int_{-\infty}^{\infty} \frac{\partial}{\partial \theta} f(x, \theta) dx$$

But when is swapping $\frac{d}{d\theta}$ and $\int$ legal?

Situation 1: Computing moments

For $\text{Exp}(\lambda)$, $EX^n = \int_0^{\infty} x^n \frac{1}{\lambda} e^{-x/\lambda} dx$.

Key observation:

$$\frac{\partial}{\partial \lambda} \left(\frac{1}{\lambda} e^{-x/\lambda} \right) = \frac{1}{\lambda^2} \left(\frac{x}{\lambda} - 1 \right) e^{-x/\lambda}$$

Situation 2: MGF $\Rightarrow$ moments

We stated in §2.3: $\frac{d}{dt} M_X(t) = EXe^{tX}$.

This requires:

$$\frac{d}{dt} \int e^{tx} f_X(x) dx = \int \frac{\partial}{\partial t} e^{tx} f_X(x) dx$$

Without this swap, the entire moment-generating theory breaks down!

Situation 3: Statistical inference

Score equations, Fisher information, and many MLE calculations require differentiating through integrals or sums. Leibniz's Rule makes this rigorous.

When Things Go Wrong — A Cautionary Example

Warning: swapping is NOT always valid

Define $f(x, \theta) = \theta e^{-\theta x}$, $x > 0$, $\theta > 0$.

We know $\int_0^\infty \theta e^{-\theta x} dx = 1$ for all θ .

Left-hand side: $\frac{d}{d\theta} 1 = 0$.

Naive right-hand side: $\int_0^\infty \frac{\partial}{\partial \theta} (\theta e^{-\theta x}) dx = \int_0^\infty (1 - \theta x) e^{-\theta x} dx = 1 - 1 = 0$.

It works here! But consider $f(x, n) = n \cdot \mathbf{1}_{[0, 1/n]}(x)$. The integral is always 1, but the pointwise limit $\frac{\partial}{\partial n} f$ has integral $\neq 0$ at the boundary — illustrating that we need **conditions**.

The key question

We need a condition that **controls the derivative uniformly** across all x , preventing the boundary behaviour from disrupting the interchange. This is the role of the **dominating function** $g(x)$.

Theorem 2.4.1 — Leibniz's Rule

Theorem 2.4.1 (CB): Leibniz's Rule

If $f(x, \theta)$, $a(\theta)$, and $b(\theta)$ are all differentiable with respect to θ , then

$$\frac{d}{d\theta} \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx = f(b(\theta), \theta) \frac{d}{d\theta} b(\theta) - f(a(\theta), \theta) \frac{d}{d\theta} a(\theta) + \int_{a(\theta)}^{b(\theta)} \frac{\partial}{\partial \theta} f(x, \theta) dx$$

Special case: constant limits

If a and b are constants (not functions of θ), the first two terms vanish:

$$\frac{d}{d\theta} \int_a^b f(x, \theta) dx = \int_a^b \frac{\partial}{\partial \theta} f(x, \theta) dx$$

This is the most common case in statistics.

Leibniz's Rule — Worked Example

Example: Differentiating a CDF

Let $F(t) = \int_0^t x^2 dx = \frac{t^3}{3}$.

Direct differentiation: $F'(t) = t^2$.

Leibniz's Rule:

$$\frac{d}{dt} \int_0^t x^2 dx = \underbrace{t^2 \cdot 1}_{f(b(\theta), \theta) \cdot b'(\theta)} - \underbrace{0^2 \cdot 0}_{f(a(\theta), \theta) \cdot a'(\theta)} + \underbrace{\int_0^t 0 dx}_{\partial f / \partial t = 0} = t^2. \checkmark$$

Statistical application: score function

Let $\ell(\theta) = \int p(x, \theta) dx = 1$ (a pdf integrates to 1). Differentiating both sides w.r.t. θ (assuming constant limits):

$$0 = \int \frac{\partial}{\partial \theta} p(x, \theta) dx = \int \frac{\partial \log p}{\partial \theta} p(x, \theta) dx = \mathbb{E} \left[\frac{\partial}{\partial \theta} \log p(X, \theta) \right]$$

The **score function** has mean zero — a foundational result in maximum likelihood estimation.

Theorem 2.4.2 — Interchanging Limit and Integral

Theorem 2.4.2 (CB): Dominated Convergence

Suppose $h(x, y)$ is continuous at y_0 for each x , and there exists an integrable function $g(x)$ such that:

- (i) $|h(x, y)| \leq g(x)$ for all x and y
- (ii) $\int_{-\infty}^{\infty} g(x) dx < \infty$

Then:

$$\lim_{y \rightarrow y_0} \int_{-\infty}^{\infty} h(x, y) dx = \int_{-\infty}^{\infty} \lim_{y \rightarrow y_0} h(x, y) dx$$

Intuition: the dominating function $g(x)$

The function $g(x)$ **controls** how large $h(x, y)$ can get. If we can bound h by an integrable function regardless of y , then the integral behaves “continuously” in y , and the limit can move inside. This is a corollary of Lebesgue’s Dominated Convergence Theorem.

Theorem 2.4.3 — Differentiating Under an Infinite Integral

Theorem 2.4.3 (CB): Differentiation Under Integral Sign

Suppose $f(x, \theta)$ is differentiable at $\theta = \theta_0$ for every x , and there exists $g(x, \theta_0)$ and $\delta_0 > 0$ such that:

$$\textcircled{i} \quad \left| \frac{f(x, \theta_0 + \delta) - f(x, \theta_0)}{\delta} \right| \leq g(x, \theta_0) \text{ for all } x \text{ and } |\delta| \leq \delta_0$$

$$\textcircled{ii} \quad \int_{-\infty}^{\infty} g(x, \theta_0) dx < \infty$$

Then:

$$\left. \frac{d}{d\theta} \int_{-\infty}^{\infty} f(x, \theta) dx \right|_{\theta=\theta_0} = \int_{-\infty}^{\infty} \left. \frac{\partial}{\partial \theta} f(x, \theta) \right|_{\theta=\theta_0} dx \quad (2.4.2)$$

Corollary 2.4.4 (practical version)

If $f(x, \theta)$ is differentiable in θ and there exists $g(x, \theta)$ with $\left| \frac{\partial}{\partial \theta} f(x, \theta) \right| \leq g(x, \theta)$ and $\int g(x, \theta) dx < \infty$, then (2.4.3) holds.

Example 2.4.5 — Interchanging $\frac{d}{d\lambda}$ and $\int$ (Part 1)

Setup: $X \sim \text{Exp}(\lambda)$, compute $\frac{d}{d\lambda} \mathbb{E}X^n$

$f_X(x) = \frac{1}{\lambda} e^{-x/\lambda}$. For integer $n > 0$:

$$\frac{d}{d\lambda} \mathbb{E}X^n = \frac{d}{d\lambda} \int_0^\infty x^n \cdot \frac{1}{\lambda} e^{-x/\lambda} dx$$

Justification (Cor. 2.4.4): The derivative of the integrand is bounded by an integrable dominating function for λ near any fixed $\lambda_0 > 0$. So we may move $\frac{d}{d\lambda}$ inside:

$$\frac{d}{d\lambda} \mathbb{E}X^n = \int_0^\infty x^n \cdot \frac{\partial}{\partial \lambda} \left(\frac{1}{\lambda} e^{-x/\lambda} \right) dx$$

Compute the partial derivative:

$$\frac{\partial}{\partial \lambda} \left(\frac{1}{\lambda} e^{-x/\lambda} \right) = \frac{x}{\lambda^3} e^{-x/\lambda} - \frac{1}{\lambda^2} e^{-x/\lambda} = \frac{e^{-x/\lambda}}{\lambda^2} \left(\frac{x}{\lambda} - 1 \right)$$

Example 2.4.5 — Exponential Moments Recursion (Part 2)

Continuing the calculation

Substituting into the integral:

$$\begin{aligned}\frac{d}{d\lambda} EX^n &= \int_0^\infty x^n \cdot \frac{1}{\lambda^2} \left(\frac{x}{\lambda} - 1 \right) e^{-x/\lambda} dx \\ &= \frac{1}{\lambda^2} \int_0^\infty \frac{x^{n+1}}{\lambda} e^{-x/\lambda} dx - \frac{1}{\lambda^2} \int_0^\infty x^n e^{-x/\lambda} dx \\ &= \frac{1}{\lambda^2} EX^{n+1} - \frac{1}{\lambda} EX^n\end{aligned}$$

Rearranging:

$$EX^{n+1} = \lambda EX^n + \lambda^2 \frac{d}{d\lambda} EX^n \quad (2.4.7)$$

Power of this recursion

Starting from $EX^0 = 1$ and $EX = \lambda$:

$$EX^2 = \lambda \cdot \lambda + \lambda^2 \cdot 1 = 2\lambda^2, \quad EX^3 = \lambda \cdot 2\lambda^2 + \lambda^2 \cdot 4\lambda = 6\lambda^3, \quad \dots$$

In general: $EX^n = n! \lambda^n$ — all moments from one recursion, no new integration!

Example 2.4.6 — Justifying $M'_X(t) = E[Xe^{tX}]$ for Normal

$$M_X(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-(x-\mu)^2/2} dx.$$

Need to move $\frac{d}{dt}$ inside. Find $g(x, t)$ with:

$$\left| \frac{\partial}{\partial t} e^{tx} e^{-(x-\mu)^2/2} \right| = |x| e^{tx} e^{-(x-\mu)^2/2} \leq g(x, t)$$

For $|t' - t| \leq \delta_0$, choose:

$$g(x, t) = |x| e^{(t+\delta_0)|x|} e^{-(x-\mu)^2/2}$$

Completing the square shows $\int g(x, t) dx < \infty$. Therefore, by Corollary 2.4.4:

$$\frac{d}{dt} M_X(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{tx} e^{-(x-\mu)^2/2} dx = E[Xe^{tX}]$$

Setting $t = 0$: $M'_X(0) = EX = \mu$. ✓

Example 2.4.7 — Computing EX for Geometric by Differentiation

$X \sim \text{Geometric}(\theta)$: $P(X = x) = \theta(1 - \theta)^x$, $x = 0, 1, 2, \dots$

We know $\sum_{x=0}^{\infty} \theta(1 - \theta)^x = 1$ for $0 < \theta < 1$.

Differentiate both sides w.r.t. θ (justified by Thm 2.4.8 — the derivative series converges uniformly):

$$\frac{d}{d\theta} \sum_{x=0}^{\infty} \theta(1 - \theta)^x = \sum_{x=0}^{\infty} \frac{d}{d\theta} \theta(1 - \theta)^x = \sum_{x=0}^{\infty} [(1 - \theta)^x - \theta x(1 - \theta)^{x-1}] = 0$$

$$\Rightarrow \underbrace{\frac{1}{\theta} \sum_{x=0}^{\infty} \theta(1 - \theta)^x}_{=1/\theta} - \underbrace{\frac{1}{1 - \theta} \sum_{x=0}^{\infty} x\theta(1 - \theta)^x}_{=EX/(1-\theta)} = 0$$

$$\Rightarrow EX = \frac{1 - \theta}{\theta}$$

Theorem 2.4.8 — Differentiating a Sum

Theorem 2.4.8 (CB): Differentiation Under a Sum

Suppose $\sum_{x=0}^{\infty} h(\theta, x)$ converges for all $\theta \in (a, b)$, and:

- Ⓐ $\frac{\partial}{\partial \theta} h(\theta, x)$ is continuous in θ for each x
- Ⓑ $\sum_{x=0}^{\infty} \frac{\partial}{\partial \theta} h(\theta, x)$ converges **uniformly** on every closed bounded subinterval of (a, b)

Then:

$$\frac{d}{d\theta} \sum_{x=0}^{\infty} h(\theta, x) = \sum_{x=0}^{\infty} \frac{\partial}{\partial \theta} h(\theta, x)$$

Key condition: uniform convergence

“Uniform convergence” means the series converges at the same rate for all θ in the interval — no particular θ causes the tail to be abnormally large.

Theorem 2.4.10 (CB)

If $\sum h(\theta, x)$ converges uniformly on $[a, b]$ and $h(\theta, x)$ is continuous in θ for each x , then:

$$\int_a^b \sum_{x=0}^{\infty} h(\theta, x) d\theta = \sum_{x=0}^{\infty} \int_a^b h(\theta, x) d\theta$$

Section 2.4 — Summary

Operation	Condition needed	Result
$\frac{d}{d\theta} \int_{a(\theta)}^{b(\theta)} f \, dx$	$f, a(\theta), b(\theta)$ differentiable	Leibniz Rule (Thm 2.4.1)
$\lim_{y \rightarrow y_0} \int h(x, y) \, dx$	$ h \leq g(x), \int g < \infty$	Thm 2.4.2
$\frac{d}{d\theta} \int_{-\infty}^{\infty} f(x, \theta) \, dx$	Dominated difference quotients, $\int g < \infty$	Thm 2.4.3 / Cor 2.4.4
$\frac{d}{d\theta} \sum_x h(\theta, x)$	Uniform convergence of $\sum \partial_\theta h$	Thm 2.4.8
$\int \sum_x h(\theta, x) \, d\theta$	Uniform convergence of $\sum h$	Thm 2.4.10

The overarching principle

Swapping $\frac{d}{d\theta}$ (or $\lim$) with $\int$ (or $\sum$) is valid when the integrand (or summand) is **uniformly controlled** by an integrable dominating function. This is the Dominated Convergence Theorem in disguise.

In-Class Activity — Section 2.4 (Q1 & Q2)

Q1 — Leibniz's Rule

Let $F(\theta) = \int_0^\theta e^{-\theta x} dx$.

- (a) Evaluate $F(\theta)$ directly (compute the integral first).
- (b) Apply Leibniz's Rule to find $F'(\theta)$.
- (c) Differentiate your answer from (a) and verify it matches (b).

Q2 — Differentiating Under the Integral (Exponential)

For $X \sim \text{Exp}(\lambda)$, we know $EX = \lambda$ and $EX^2 = 2\lambda^2$.

- (a) Use the recursion (2.4.7): $EX^{n+1} = \lambda EX^n + \lambda^2 \frac{d}{d\lambda} EX^n$ to find EX^3 without integration.
- (b) Verify using the general formula $EX^n = n! \lambda^n$.

In-Class Activity — Section 2.4 (Q3)

Q3 — Score Function Has Mean Zero

Let $f(x, \theta)$ be a pdf with $\int_{-\infty}^{\infty} f(x, \theta) dx = 1$.

Assume the interchange of differentiation and integration is valid.

(a) Differentiate both sides w.r.t. θ to show:

$$\int_{-\infty}^{\infty} \frac{\partial}{\partial \theta} \log f(x, \theta) \cdot f(x, \theta) dx = 0$$

(b) Interpret this result: what does it say about $E\left[\frac{\partial}{\partial \theta} \log f(X, \theta)\right]$?

Q4 — Geometric Mean by Differentiation

For $X \sim \text{Geometric}(\theta)$: $P(X = x) = \theta(1 - \theta)^x$, $x = 0, 1, 2, \dots$

(a) Start from $\sum_{x=0}^{\infty} \theta(1 - \theta)^x = 1$.

(b) Differentiate once w.r.t. θ (term by term) and rearrange to confirm $EX = \frac{1 - \theta}{\theta}$.

(c) Differentiate *again* to find EX^2 , then compute $\text{Var}(X)$.

Activity Solutions — Q1 & Q2

(a) $F(\theta) = \int_0^\theta e^{-\theta x} dx = \left[-\frac{1}{\theta} e^{-\theta x}\right]_0^\theta = \frac{1}{\theta}(1 - e^{-\theta^2}).$

(b) Leibniz: $F'(\theta) = e^{-\theta^2} \cdot 1 - e^0 \cdot 0 + \int_0^\theta (-x)e^{-\theta x} dx = e^{-\theta^2} - E[X \cdot \mathbf{1}_{X \leq \theta}]$
 $= e^{-\theta^2} - \int_0^\theta x e^{-\theta x} dx.$

(c) Differentiate (a): $F'(\theta) = \frac{d}{d\theta} \left[\frac{1 - e^{-\theta^2}}{\theta} \right] = \frac{2e^{-\theta^2}}{\theta} - \frac{1 - e^{-\theta^2}}{\theta^2}.$ Both (b) and (c) give the same expression. ✓

(a) $EX^3 = \lambda \cdot EX^2 + \lambda^2 \frac{d}{d\lambda} EX^2 = \lambda \cdot 2\lambda^2 + \lambda^2 \cdot 4\lambda = 2\lambda^3 + 4\lambda^3 = 6\lambda^3.$

(b) General formula: $EX^3 = 3! \lambda^3 = 6\lambda^3.$ ✓

Activity Solutions — Q3 & Q4

(a) Differentiate $\int f(x, \theta) dx = 1$: $0 = \int \frac{\partial}{\partial \theta} f(x, \theta) dx = \int \frac{\partial \log f}{\partial \theta} \cdot f(x, \theta) dx$.

(b) This equals $E\left[\frac{\partial}{\partial \theta} \log f(X, \theta)\right] = 0$.

The **score function** $\frac{\partial}{\partial \theta} \log f(X, \theta)$ has mean zero under the true parameter θ — a fundamental fact in MLE theory.

(b) $\frac{d}{d\theta} 1 = 0 = \sum_{x=0}^{\infty} [(1-\theta)^x - x\theta(1-\theta)^{x-1}] = \frac{1}{\theta} - \frac{EX}{1-\theta}$. So $EX = \frac{1-\theta}{\theta}$. ✓

(c) Differentiate again: $\frac{d^2}{d\theta^2} 1 = 0$. After algebra: $EX^2 = \frac{(1-\theta)(2-\theta)}{\theta^2}$.

$\text{Var}(X) = EX^2 - (EX)^2 = \frac{(1-\theta)(2-\theta)}{\theta^2} - \frac{(1-\theta)^2}{\theta^2} = \frac{1-\theta}{\theta^2}$.